Laboratory Report 1

Software Relevant Tools, Standards, and Code Versioning using Github

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Program:	Computer Engineering	**Section:**	B3

A. Learning Outcomes:

At the end of this laboratory, students should be able to:

1. Identify software development tools and standards used in software projects.
2. Create and manage a GitHub repository.
3. Apply Git version control commands.
4. Collaborate using GitHub features.
5. Demonstrate proper documentation and coding standards.

B. Laboratory Scenario:

You have been hired as a junior software developer in a software company. Your team uses GitHub for source code management and follows coding standards to maintain software quality. Your task is to create a project repository, manage versions of source code, and document your work according to industry standards.

C. Practical Tasks:

1. Software Development Tools and Standards

Create a document named Tools_and_Standards.md containing:

a. List at least 5 development tools.
b. Describe the purpose of each tool.
c. Identify at least three coding standards or best practices used in software development.

(Please include the following: **1.** Screenshots of the output)

Screenshot of Output
Tools_and_Standards.md - Notepad File Edit Format View Help 1. Text Editor - Can be used for writing and editing code, where source code is first written in. (example, VS Code) 2. Terminal - Used for running commands like file directory navigation and executing code directly. (example, Bash) 3. Code Repository Host - Useful for storing project files, but more useful for collaboration purposes. (example, Github) 4. Testing Platform - Designed for testing software before publication. (example, Postman) 5. Version Tracking - Allows for taking snapshots of files periodically, ensuring there's a safe fallback when things go wrong. (example, Git)

2. GitHub Repository Creation
   a. Create a GitHub account (if none exists).
   b. Create a public repository named: Software-Versioning-Lab
   c. Add the following:
      i. README.md
      ii. LICENSE
      iii. .gitignore
   d. Upload this laboratory file in this repository once accomplished.
   e. Submit GitHub repository link
   f. Screenshot of repository homepage

Screenshot of Output

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Laboratory Report 1

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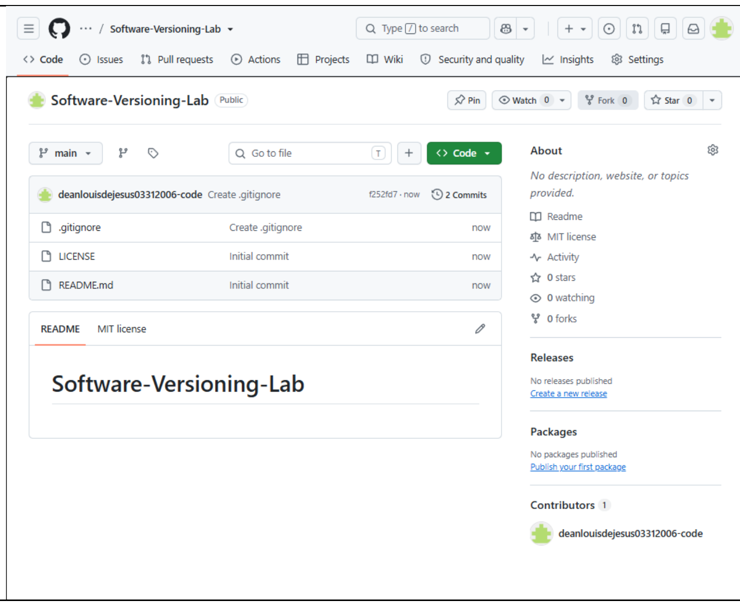

3. Git Version Control
    a. Perform the following Git commands and document them in a file named: (Git_Report.md)
        i. Clone Repository
            a) git clone <repository-url>
        ii. Creating New Branch
            a) git checkout -b feature-update
        iii. Modify README.md
            a) Add your name and laboratory information
        iv. Commit Changes
            a) git add .
            b) git commit -m "Updated README with laboratory information"
        v. Push Changes
            a) git push origin feature-update
        vi. Create Pull Request
            a) feature-update → main
    b. Screenshot the following:
        i. Commands executed
        ii. Pull Request
        iii. Commit history

(Please include the following: **1.** Screenshots of the output)

Screenshot of Output

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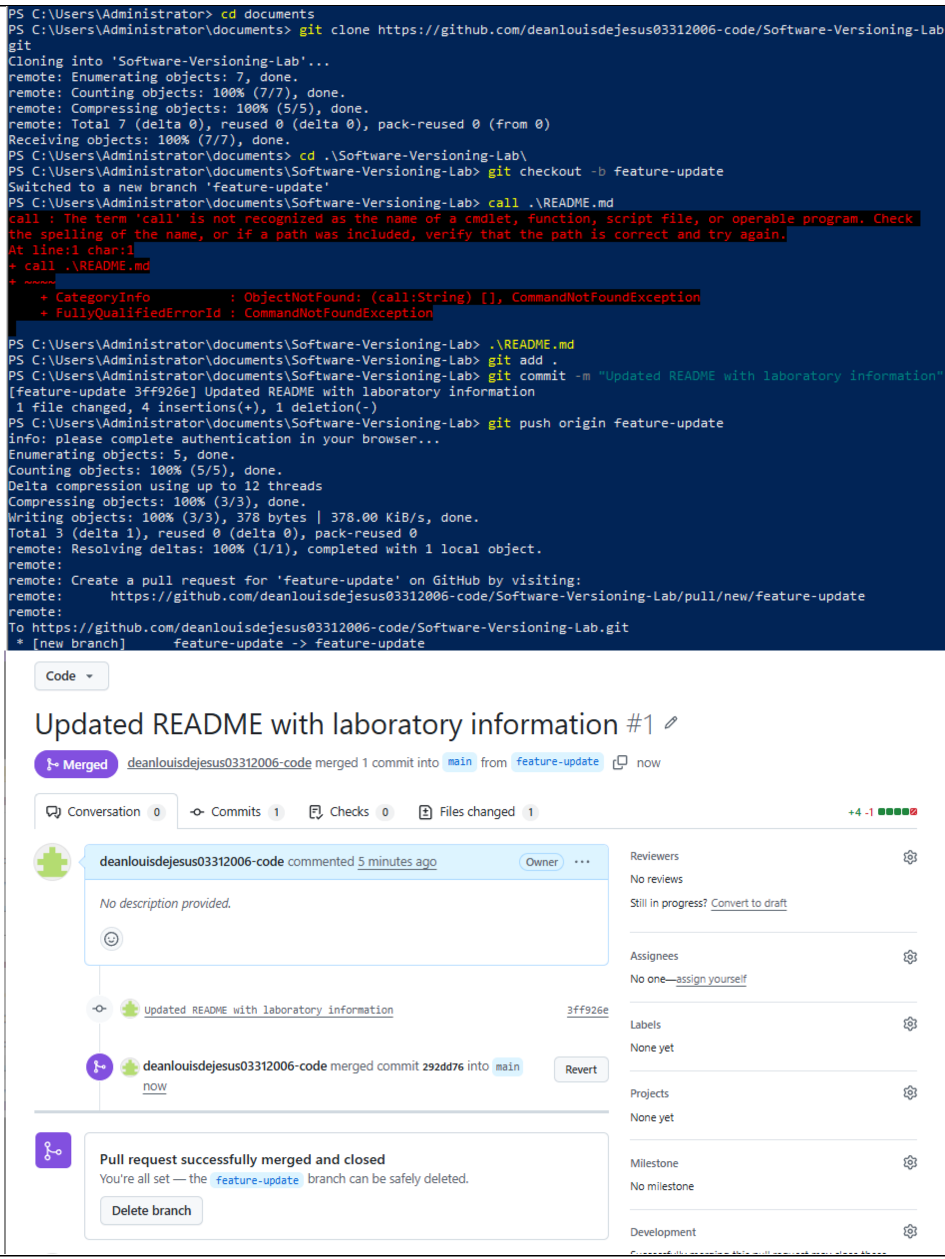

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D. Findings, Observations, and Comments:

Guide Questions:

1. What additional steps are taken in this activity?
2. Are there any difficulties during the GitHub setup?
3. What are your findings, observations, and comments in this exercise?

1. I believe that the ordering of instructions are pretty confusing, so there are many additional steps taken for this activity, such as this pretty long one for Practical Task 3B, where we need to show commit history and paste it here, while needing to put this pdf file inside the repository as well. This would require me to show proof in 3BIII that I already sent this file in the repository (as of practical task 2D), while I am still writing it.

2. Setting up GitHub wasn't too difficult at the time I made this laboratory, as I already have a github account and have git locally, from prior activities done before this course.

3. Despite already having github, its actually my first time creating my own repository and doing anything other than cloning, so I learned a lot in how to navigate github as a platform.

PICTURE/PROOF OF DOING THE ACTIVITY

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Laboratory Report 1

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E. Score Sheet

Criteria	Poor (25%)	Fair (50%)	Good (75%)	Excellent (100%)	Score
I. Completeness, Documentation, and Organization of Report	The report was incomplete, messy, and had many unanswered questions.	The report was complete, messy, and had many unanswered questions.	The report was complete, neat, and had 1 or 2 unanswered questions.	The report was complete, neat, and all the questions had been answered.	
II. Coding Design and Patterns	The program had inappropriate elements and structures, incorrect indention, and poor program documentation.	The program had corrected indention, but inappropriate elements and structures and poor program documentation.	The program had corrected indention, adequate program documentation, but inappropriate elements and structures.	The program had appropriate elements and structures, correct indention, and excellent program documentation.	
III. Findings, Observations, and Comments	The findings. observations, and comments were not based on the gathered data and results. All thoughts were inconsistent and unclear.	The findings. observations, and comments based on the gathered data and results but were not elaborated. Not all thoughts were consistent and clear.	The findings. observations, and comments were based on the gathered data and results and were mostly elaborated. Most of the thoughts were consistent and clear.	The findings. observations, and comments were based on the gathered data and results and were completely elaborated. All the thoughts were consistent and clear.	
IV. Grammar and Composition	The words used were inappropriate, wrong grammar usage, and poor sentence construction was observed.	The words used were appropriate; however, bad grammar and poor sentence construction were observed.	The words used were appropriate, had proper grammar usage, but poor sentence construction was observed.	The words used were appropriate, had proper grammar usage, and excellent sentence construction.	
SCORE:					

CHECKED			
Rating		**Signature**	
		Date	